



Politecnico
di Torino



NC STATE
UNIVERSITY

October 15-18, 2023 | Phoenix, AZ

Modeling energy storage in capacity expansion models: an analysis of the Italian energy system

Matteo Nicoli, Victor Duraes de Faria, Anderson Rodrigo de Queiroz, Laura Savoldi

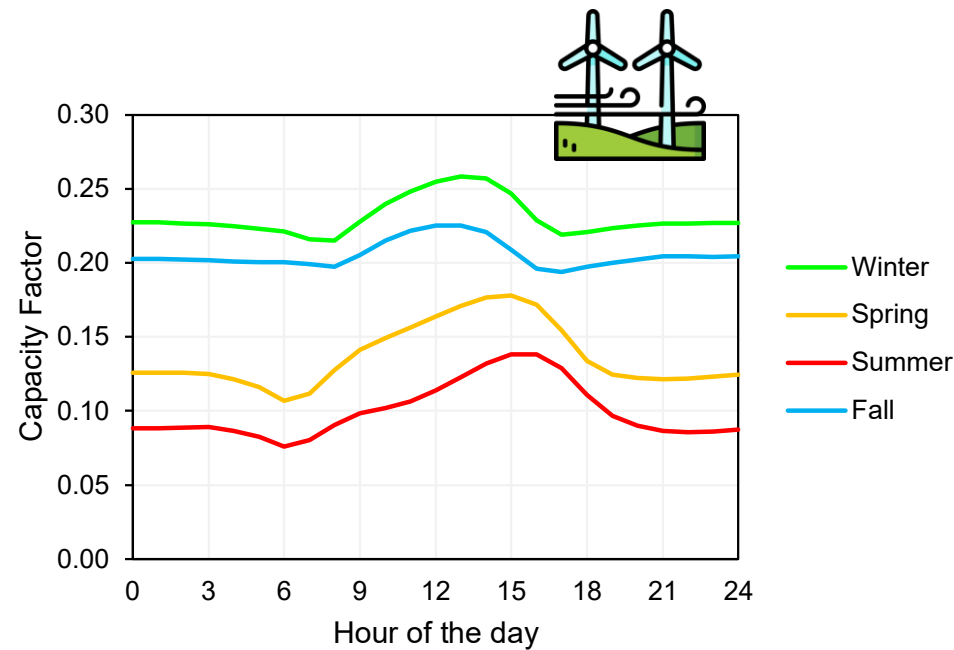
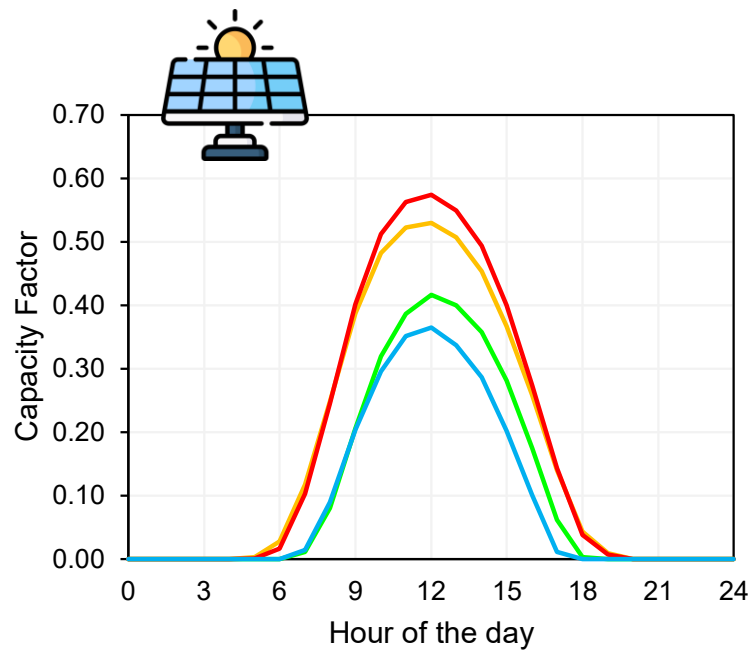
**MAHTEP Group, Energy Department "Galileo Ferraris", Politecnico di Torino
Department of Civil, Construction and Environmental Engineering, North Carolina State University**

The state of the Italian energy system



The possible role of energy storage

Renewables are a low-carbon but intermittent electricity source



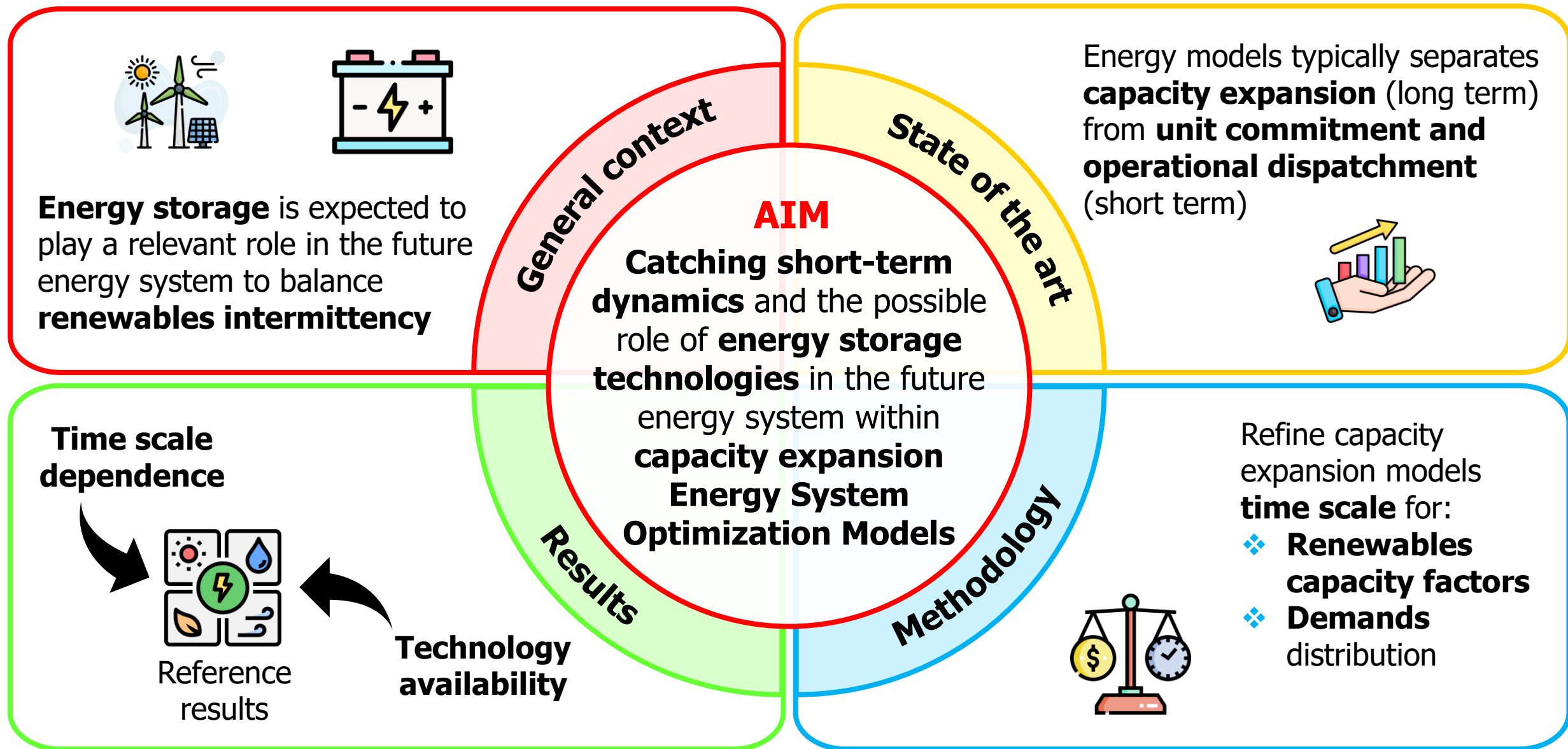
Energy storage could contribute to the energy transition in systems with a high renewable share

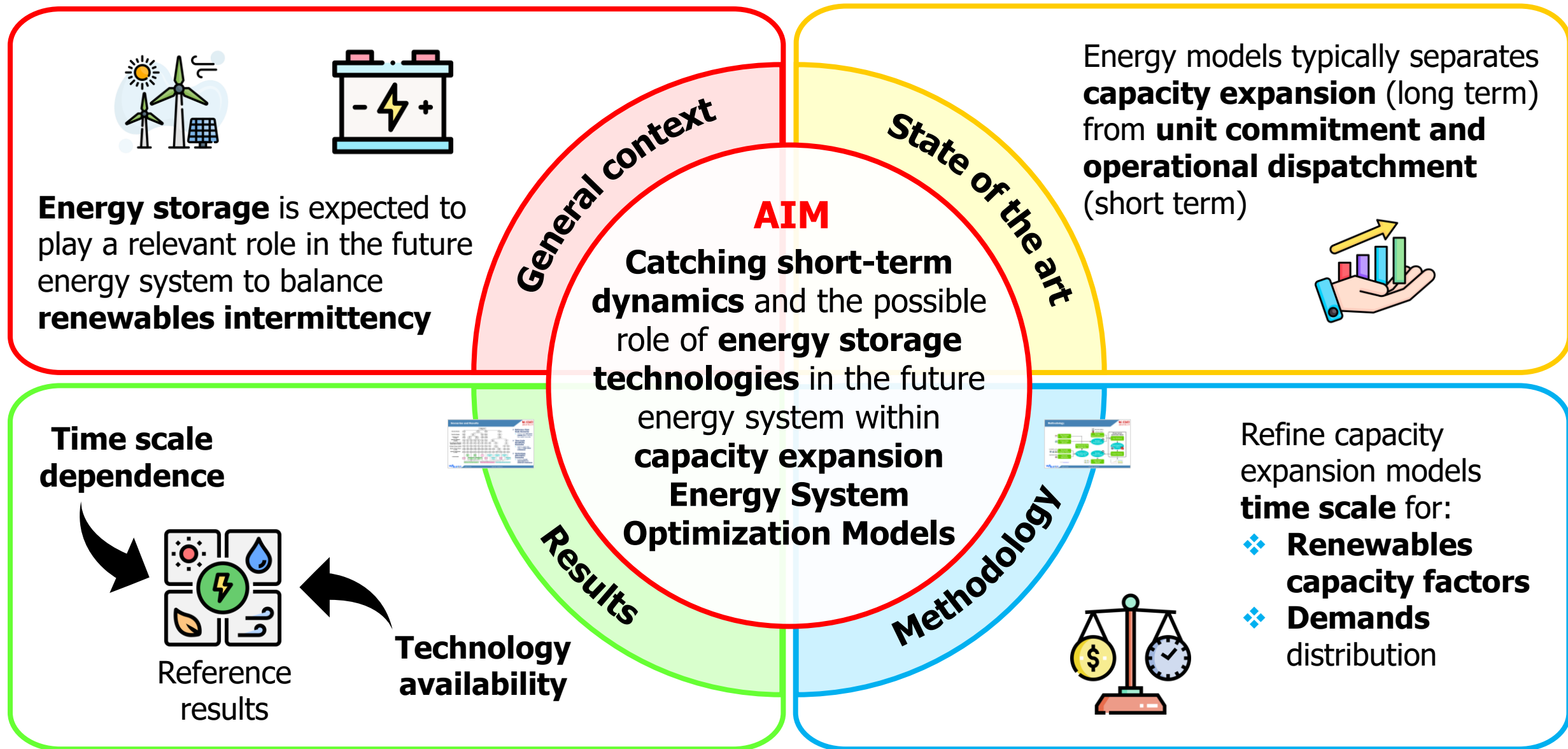
Italian official targets for 2050:

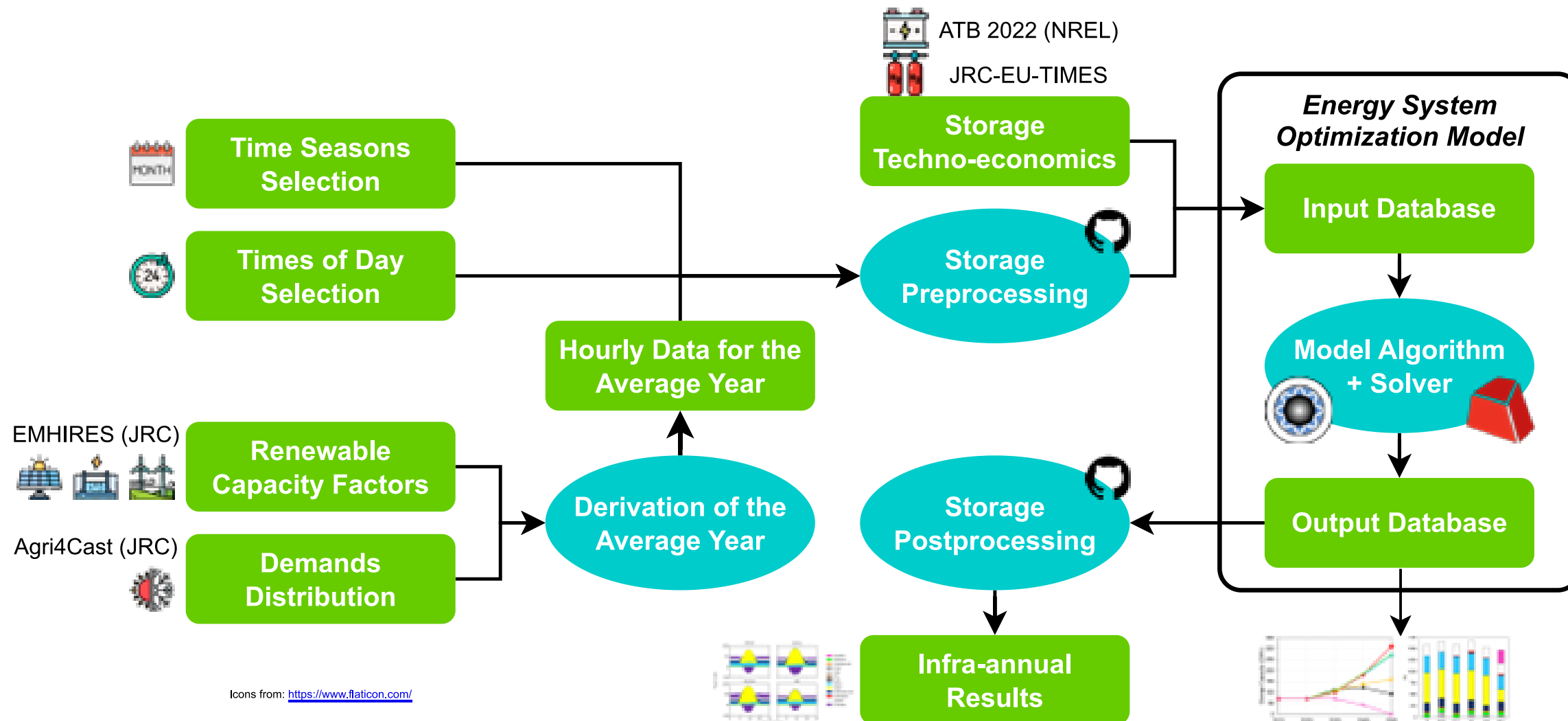
- ❖ Solar $\approx$ 200 GW
- ❖ Wind 40 ~ 50 GW
- ❖ Storage 30 ~40 GW

Solar and wind average capacity factor in Italy by season and hour of the day for the historical period 1986-2015.

Source: [EMHIRES Database by the JRC](#)







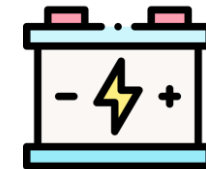
Electricity Storage Options: Lithium-Ion Batteries, Flywheels, CAES, NaS Batteries, Lead-acid Batteries, Flow Batteries, SMES, Super Capacitors

Hydrogen Storage Options: Tanks

Total system cost $\left[\frac{\$}{\text{kW}} \right] = \text{Battery Energy Cost} \left[\frac{\$}{\text{kWh}} \right] \cdot \text{Storage Duration [h]} + \text{Battery Power Cost} \left[\frac{\$}{\text{kW}} \right] + \text{Constant Cost [\$]}$

Fixed O&M cost $\left[\frac{\$}{\text{kW} \cdot \text{y}} \right] = 2.5\% \cdot \text{Total system cost} \left[\frac{\$}{\text{kW}} \right]$

Size	Lifetime (years)	Round-trip Efficiency (%)	Cost _{Power} (\$ ₂₀₂₀ /kW)		Cost _{Energy} (\$ ₂₀₂₀ /kWh)		Cost _{Constant} (\$ ₂₀₂₀)	
			2020	2050	2020	2050	2020	2050
Utility-scale	15	85	249	190	369	118		
Commercial	15	85	205	142	324	82		
Residential	15	85	655	247	685	320	6892	2963



Lithium-ion batteries

Source: ATB 2022, NREL (Moderated Scenario)

Round-trip efficiency

$$\text{Storage}_{\text{Energy},s,d} = \sum_{i,o,s,d} \text{Flow}_{i,o,s,d}^{\text{IN}} \cdot \text{Efficiency} - \sum_{i,o,s,d} \text{Flow}_{i,o,s,d}^{\text{OUT}}$$

Storage level

$$\text{Storage}_{\text{Level},s,d} = \text{Storage}_{\text{Level},s,d,\text{previous}} + \text{Storage}_{\text{Energy},s,d}$$

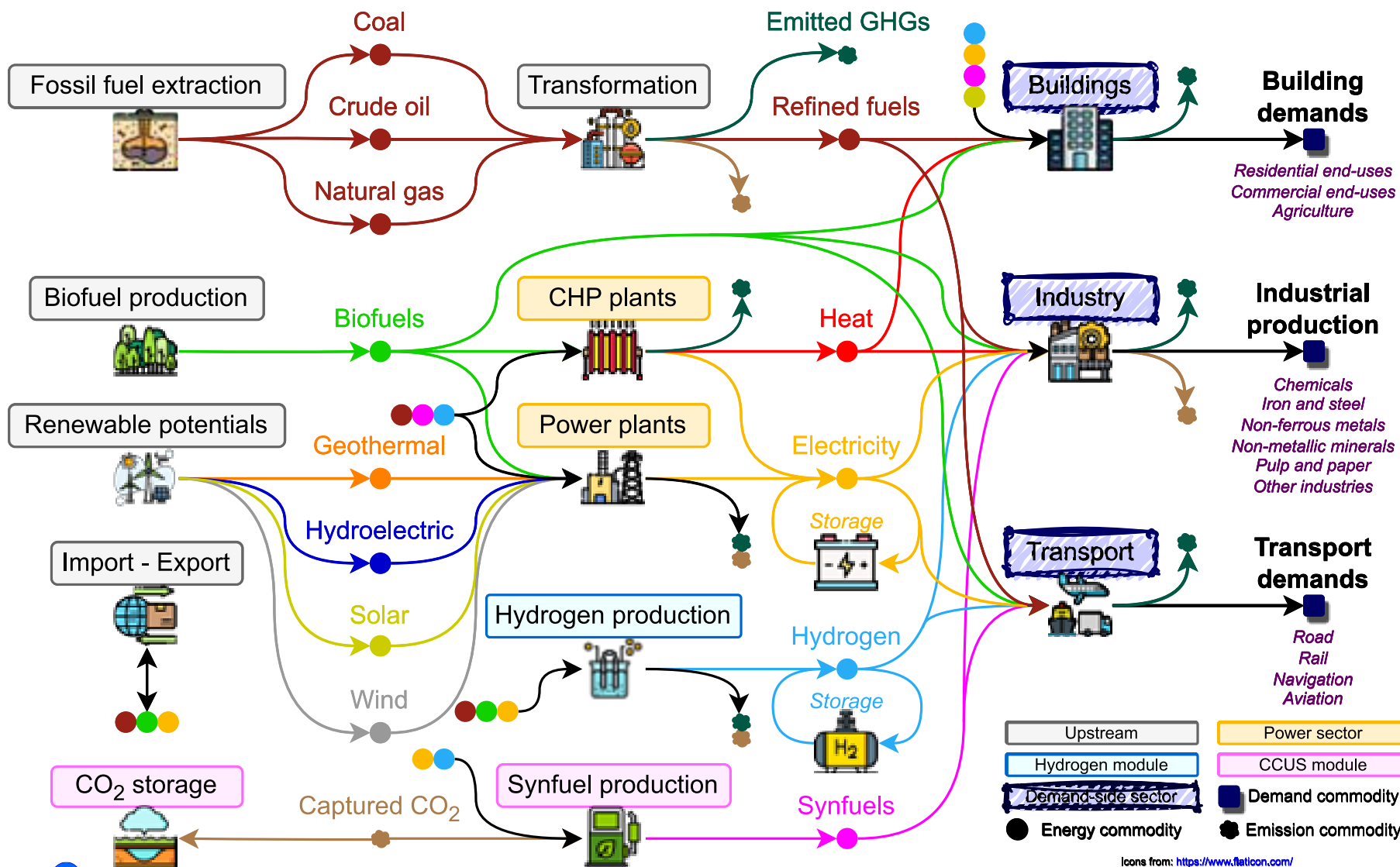
No energy exchange between different milestone years

$$\text{Storage}_{\text{Level},s,d,\text{last}} = \text{Storage}_{\text{Level},s,d,\text{first}}$$

Quantification of the storage capacity

$$\text{Storage}_{\text{Level},s,d} = \text{Cap} \cdot \text{CapToAct} \cdot \frac{\text{Storage}_{\text{Duration}}}{8760 \frac{\text{h}}{\text{y}}} \cdot \sum_d \text{SegFrac}_{s,d} \cdot 365 \frac{\text{days}}{\text{y}}$$

The TEMOA-Italy model



Energy system optimization model for the **Italian** energy system, developed within the **TEMOA** modeling framework.

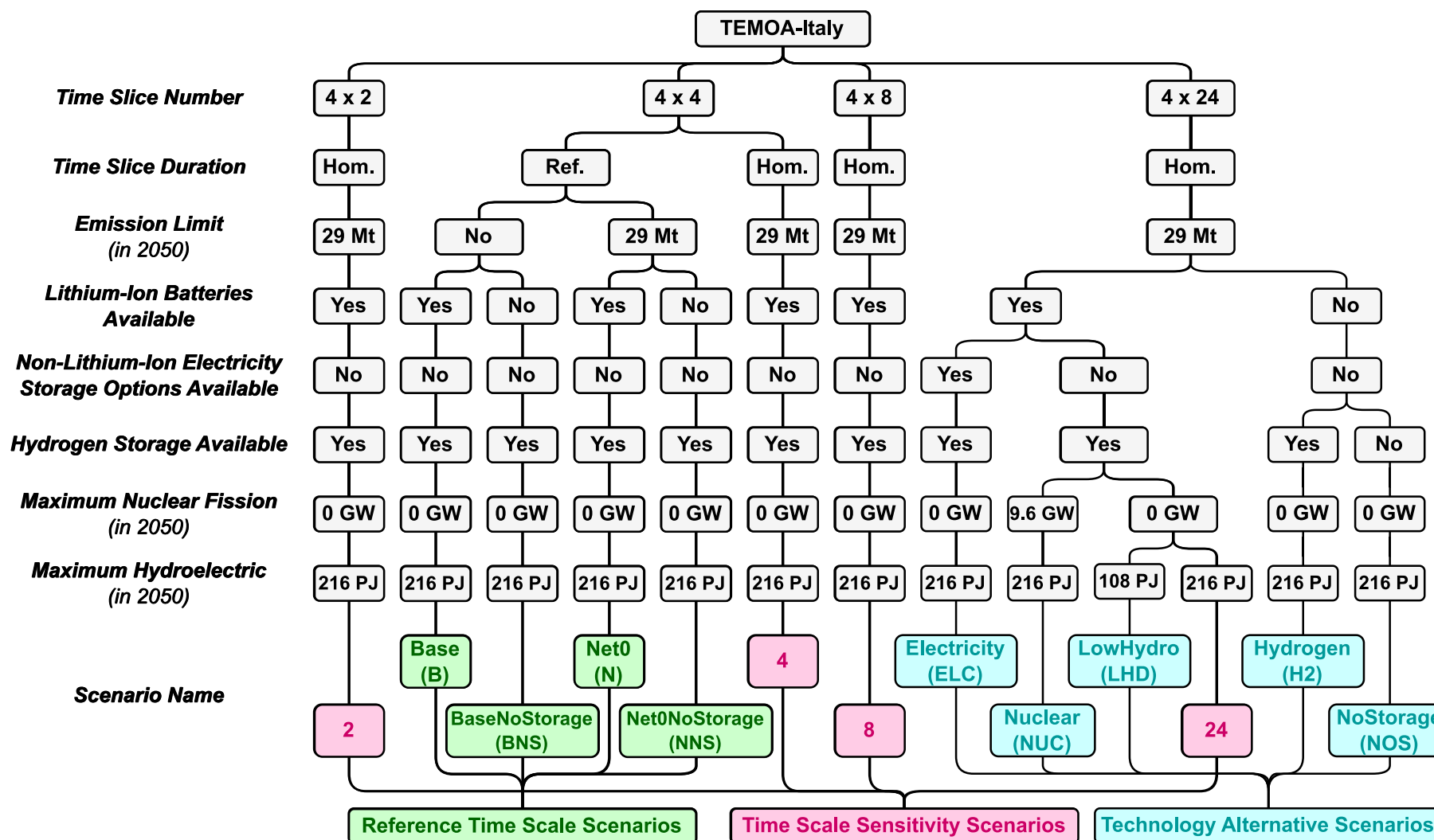
Main features:

- ❖ **Modeling all the energy sectors**
- ❖ **Oriented to capacity expansion**
- ❖ **Single region**
- ❖ **Time horizon**

2007 2010 2020
2030 2040 2050

Nicoli et al., "Can We Rely on Open-Source Energy System Optimization Models? The TEMOA-Italy Case Study", Energies, 2022

Colucci et al., "A dynamic accounting method for CO₂ emissions to assess the penetration of low-carbon fuels: application to the TEMOA-Italy energy system optimization model", Applied Energy, 2023



❖ Reference Time Scale Scenarios

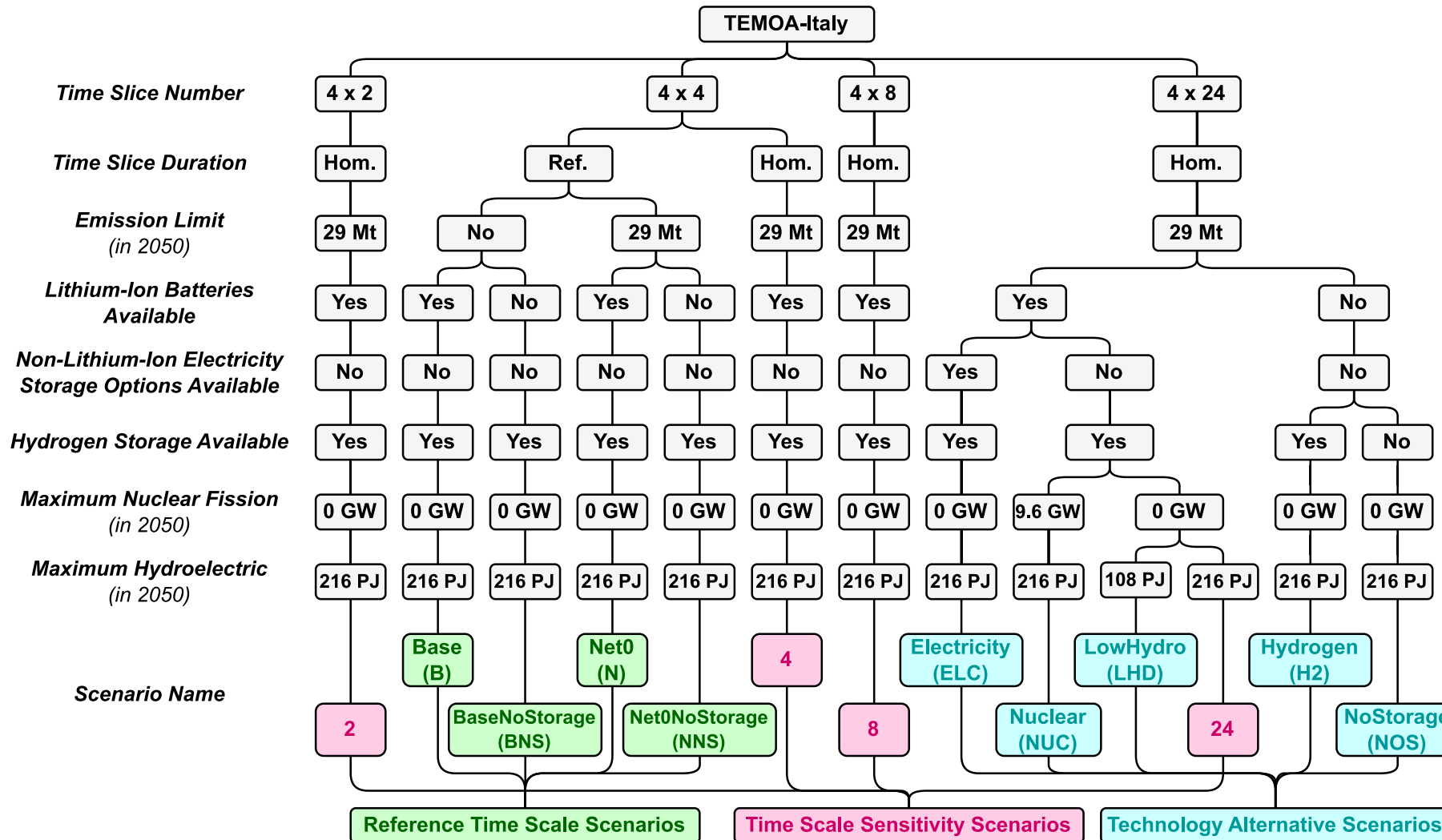
- ❖ Used to get **reference results** before refining the model time scale

❖ Time Scale Sensitivity Scenarios

- ❖ Used to assess the effect of **time slices refinement**

❖ Technology Alternative Scenarios

- ❖ Explore **results dependence on sectorial constraints**



❖ Reference Time Scale Scenarios

- ❖ Used to get **reference results** before refining the model time scale

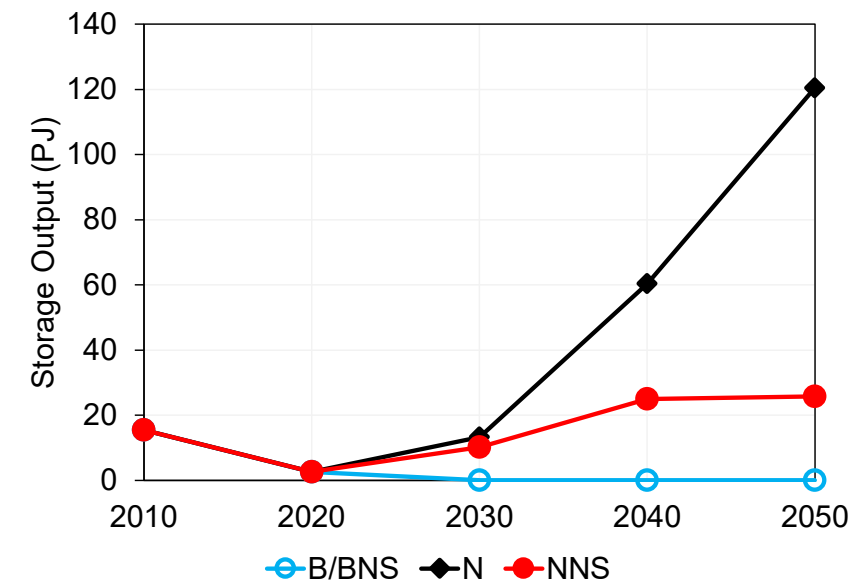
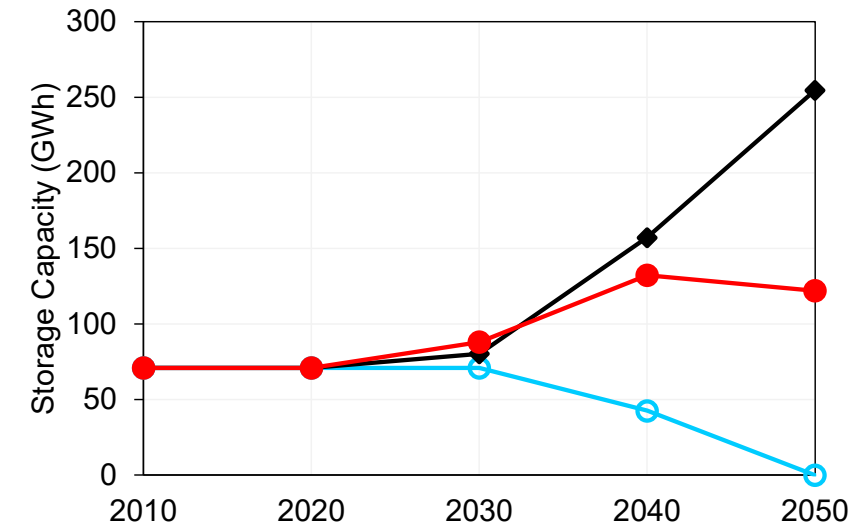
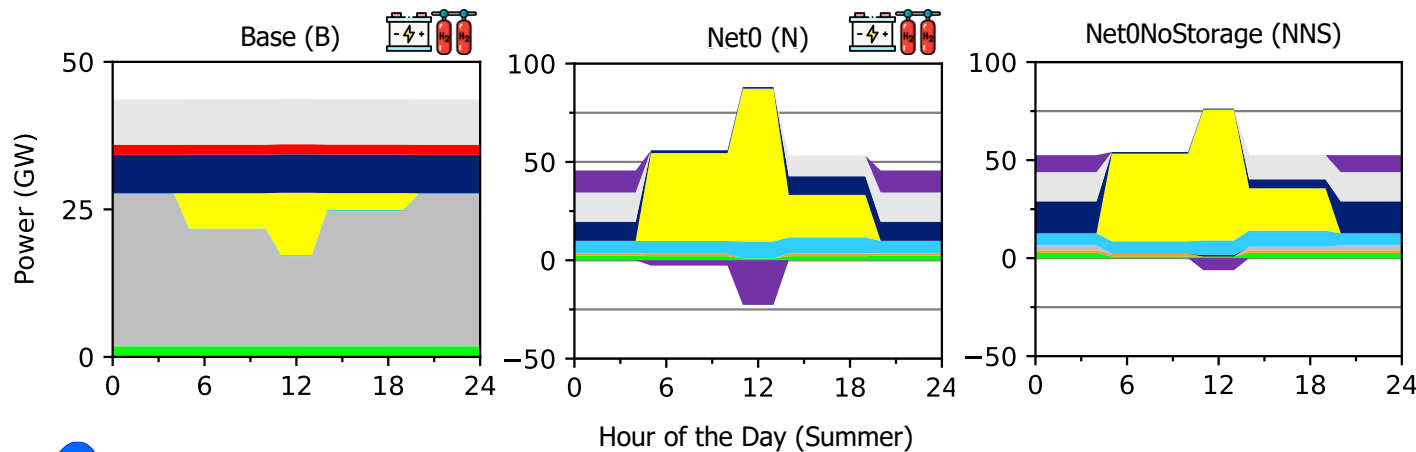
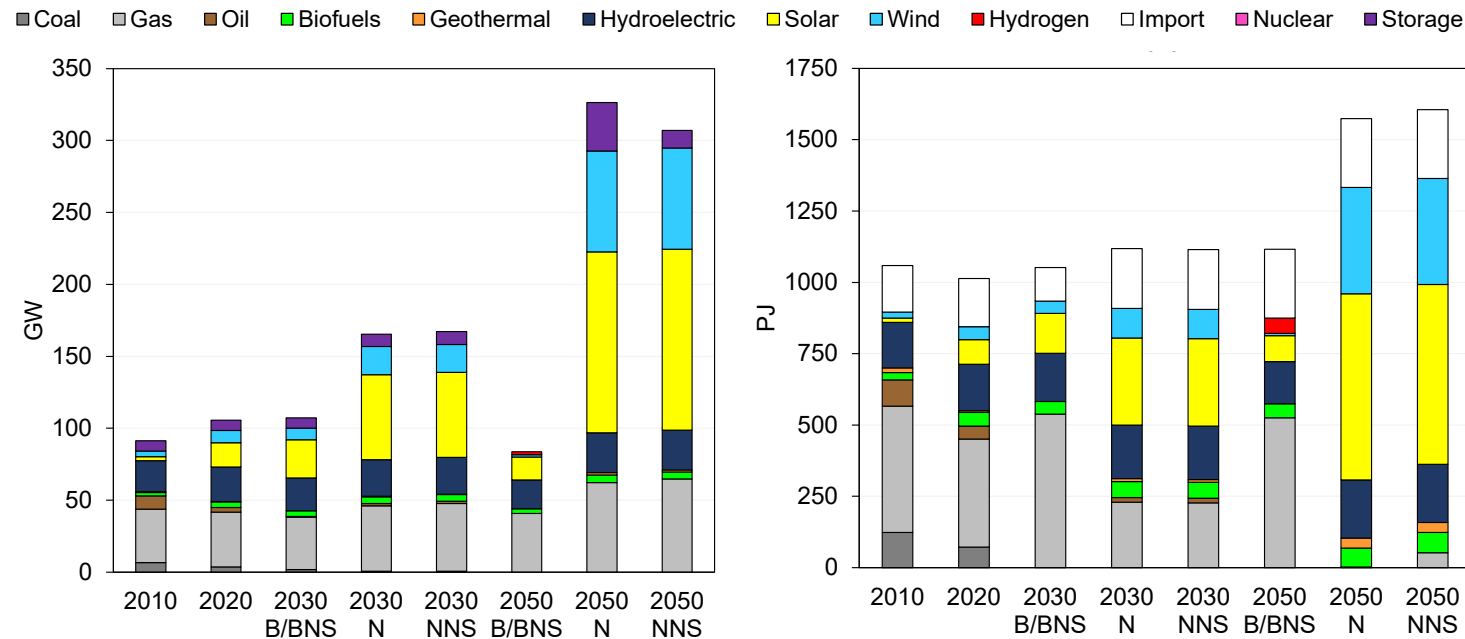
❖ Time Scale Sensitivity Scenarios

- ❖ Used to assess the effect of **time slices refinement**

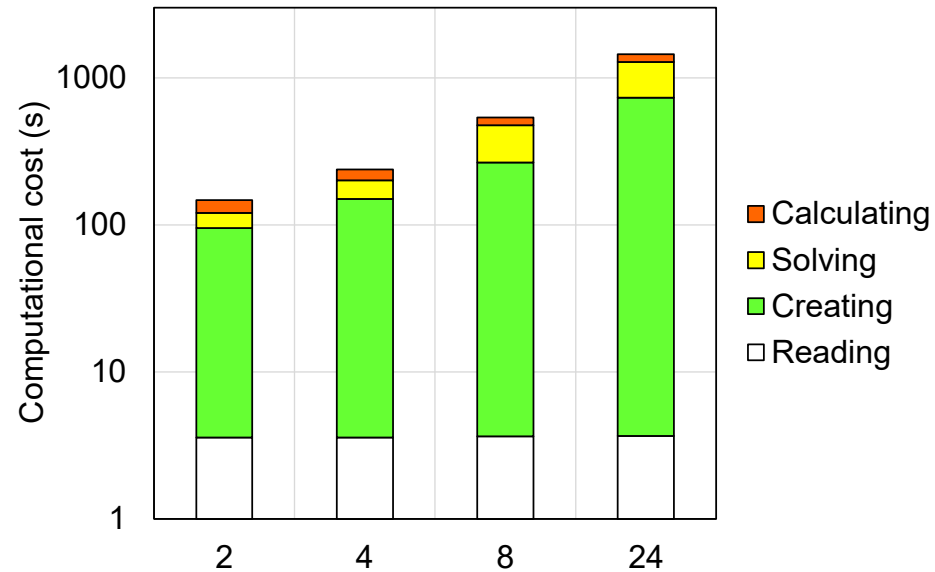
❖ Technology Alternative Scenarios

- ❖ Explore **results dependence on sectorial constraints**

Results: reference time scale scenarios



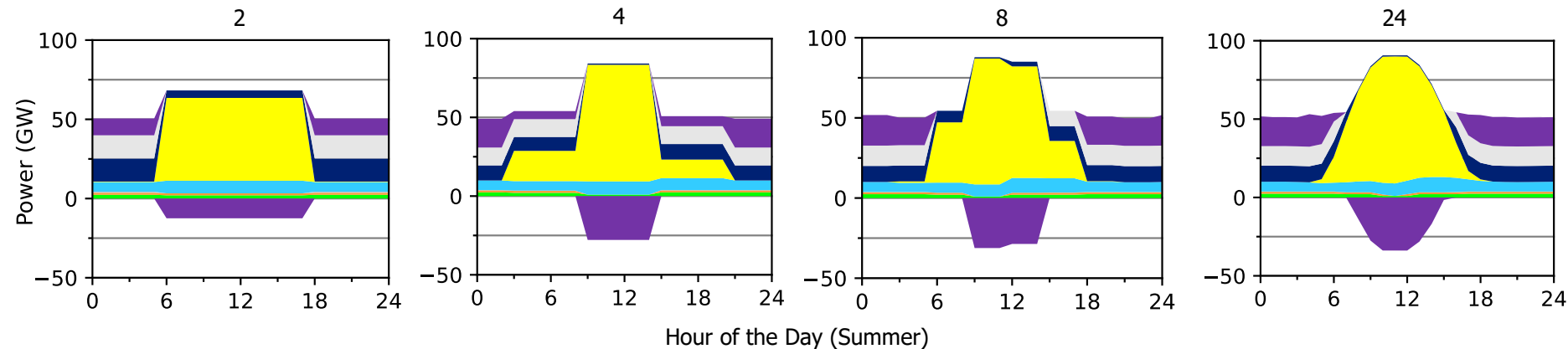
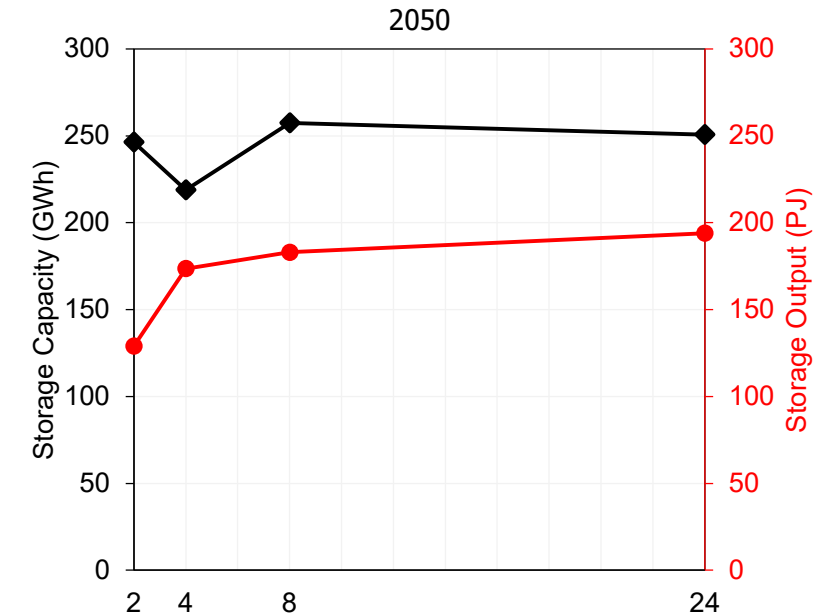
Results: time scale sensitivity scenarios



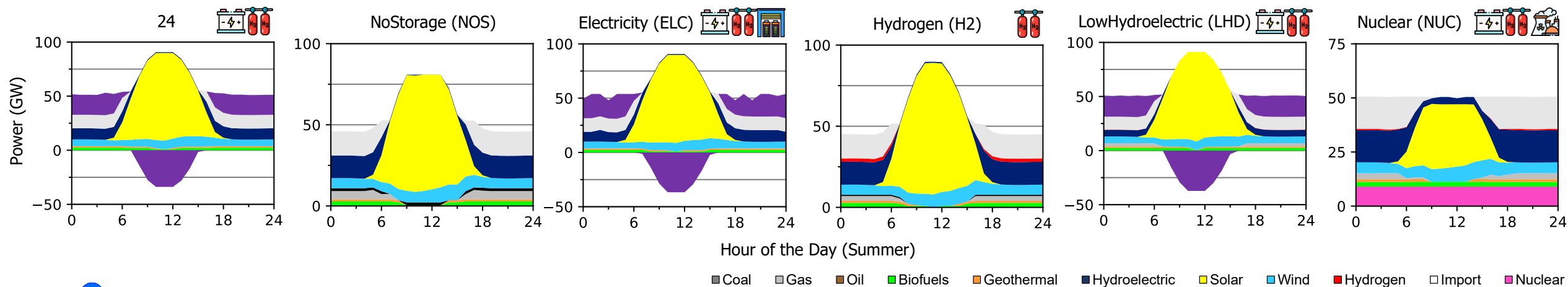
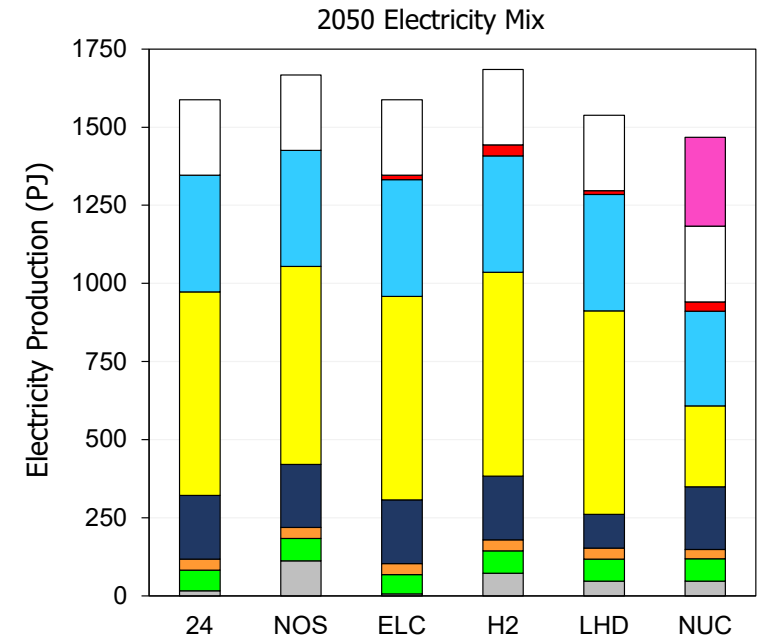
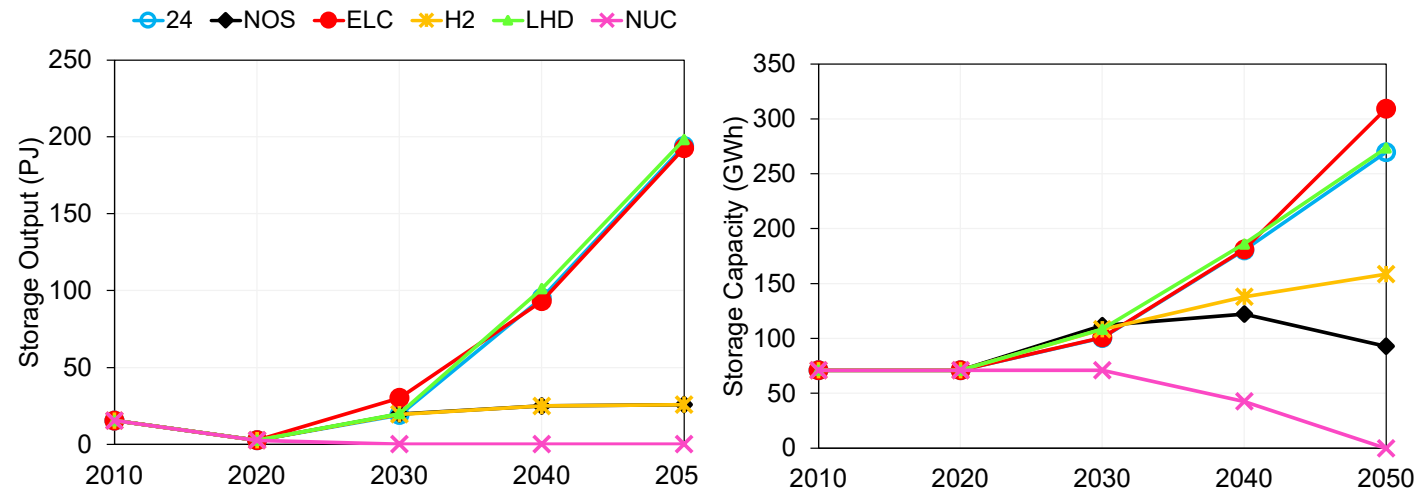
Solver: Gurobi
CPU: Intel(R) Xeon(R) Gold 6248R 3.00GHz (24 core)
Memory: 256 GB

Objective function

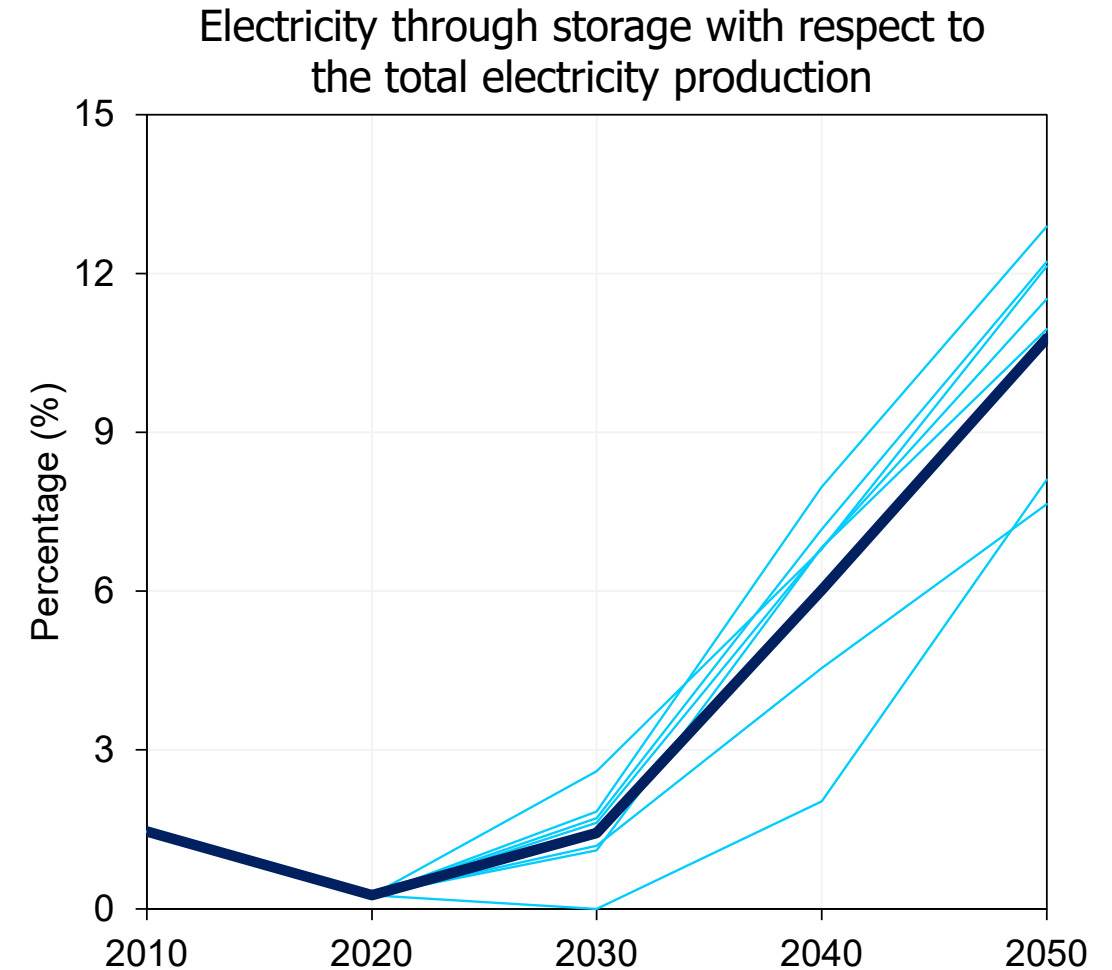
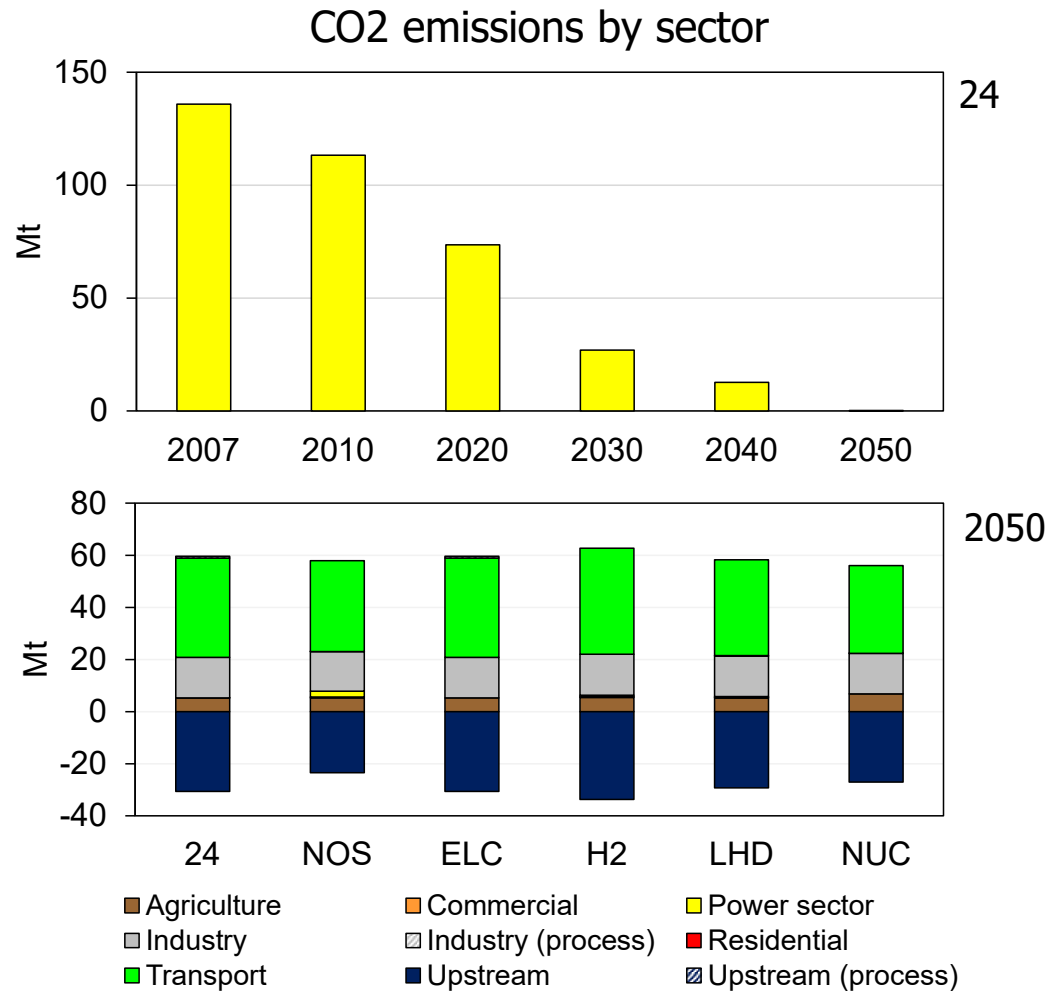
2	3.989 E6 M€
4	3.998 E6 M€
8	4.001 E6 M€
24	4.004 E6 M€



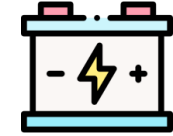
Results: technology alternative scenarios (I)



Results: technology alternative scenarios (II)



- ❖ **Energy storage plays a role** in capacity expansion models
- ❖ Energy storage **competes** with **hydroelectric and import** in satisfying the demand **during the night**
- ❖ Electricity storage **enables complete decarbonization** of the Italian power sector in 2050
- ❖ Prospects:
 - ❖ Improve the **time distribution of the demand side**
 - ❖ **Uncertainty assessment**



Thank you!



[TEMOA-Italy GitHub Repository](#)



[Let's connect on LinkedIn!](#)